

Python-Based Closed-Loop Suspended Mirror Stabilization Using Active Damping

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1. ABSTRACT

Earlier, the ability to sense mirror oscillation by processing four OSEM TIA outputs, and to actuate the mirror to oscillate at its resonant mode, was presented. This report presents Python-controlled OSEM-suspended mirror stabilization via a closed-loop velocity damping algorithm. The ability to target stabilizing a particular OSEM channel is also demonstrated. It was observed that, along with the targeted OSEM channel, all other channels were also stabilized, with the targeted channel exhibiting the best flatness. The operating Python programs are developed with Claude Sonnet-5.

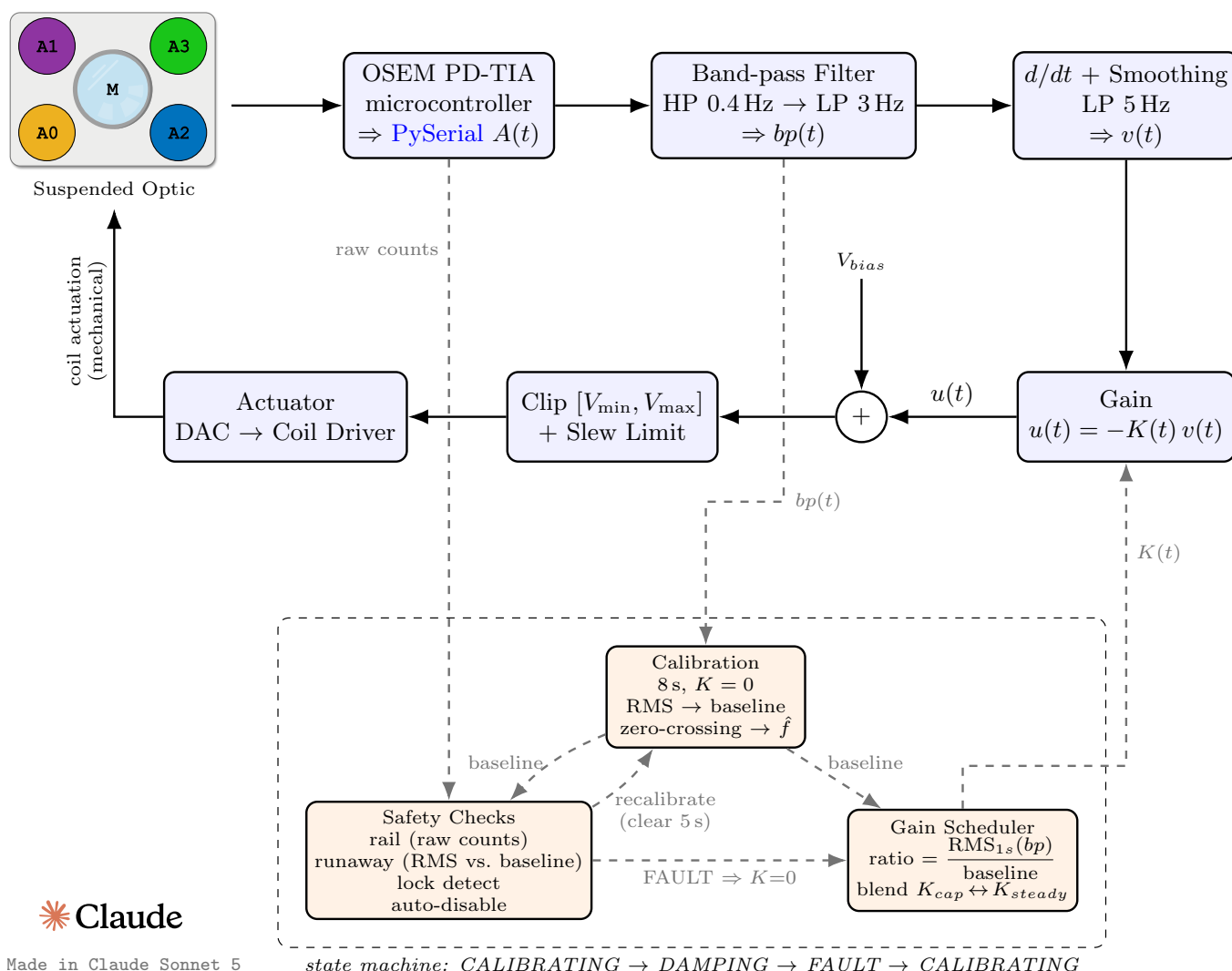
2. THE ACTIVE DAMPING ALGORITHM

Each suspended mirror is monitored by four OSEM channels (A0–A3, one per quadrant), whose photodiode–transimpedance amplifier output is digitised by a microcontroller and streamed to the host as a raw amplitude $A(t)$ per channel. Because this signal is dominated by the mirror's own rigid-body resonance, a band-pass filter first isolates that motion from DC drift and high-frequency sensor noise,

$$bp(t) = LP_{3\text{ Hz}}[HP_{0.4\text{ Hz}}[A(t)]], \quad (1)$$

Per-channel OSEM closed-loop damping algorithm

(osem_fast_lock.py – one instance runs independently per OSEM/coil pair)



Made in Claude Sonnet 5

state machine: CALIBRATING → DAMPING → FAULT → CALIBRATING

and a smoothed first-order backward-difference estimate of $bp(t)$'s derivative gives a live velocity estimate,

$$v(t) = \text{LP}_{5\text{ Hz}} \left[\frac{bp(t) - bp(t - \Delta t)}{\Delta t} \right]. \quad (2)$$

This drives a velocity-feedback, or “cold damping,” control law,

$$u(t) = -K(t)v(t), \quad (3)$$

which behaves as an artificial viscous force: pushing the coil against the mirror's current motion removes energy from the resonance every cycle. The correction $u(t)$ is added to a fixed bias, clipped to the actuator's safe range, slew-limited, and sent to the coil – closing the loop, since the same coil drives the mirror the OSEM is watching. The gain $K(t)$ is not fixed: an 8-second open-loop calibration at the start of every run measures the mirror's natural, undamped oscillation as a baseline RMS,

$$\text{RMS}_{\text{base}} = \sqrt{\frac{1}{N} \sum_{i=1}^N bp(t_i)^2}, \quad (4)$$

against which the live oscillation is continuously compared,

$$\text{ratio}(t) = \frac{\text{RMS}_{1s}[bp(t)]}{\text{RMS}_{\text{base}}}. \quad (5)$$

While this ratio is large the scheduler blends towards a stronger capture gain K_{cap} for fast pull-in; once the oscillation has largely settled it relaxes towards a gentler, validated steady-state gain K_{steady} ,

$$K(t) = K_{\text{steady}} + \text{clip} \left(\frac{\text{ratio}(t) - 0.2}{0.6 - 0.2}, 0, 1 \right) (K_{\text{cap}} - K_{\text{steady}}). \quad (6)$$

A supervisory state machine runs alongside this loop, watching for a railed sensor or a runaway oscillation; either condition forces $K \rightarrow 0$ and freezes the actuator until the channel has been clear for several seconds, at which point calibration – and damping – resume automatically. In the figure these slower interactions are drawn as dashed arrows, distinct from the solid, per-sample signal path above: raw counts and $bp(t)$ are tapped down into the calibration and safety blocks, while the resulting gain $K(t)$ is fed back up into the multiplier that closes the main loop.

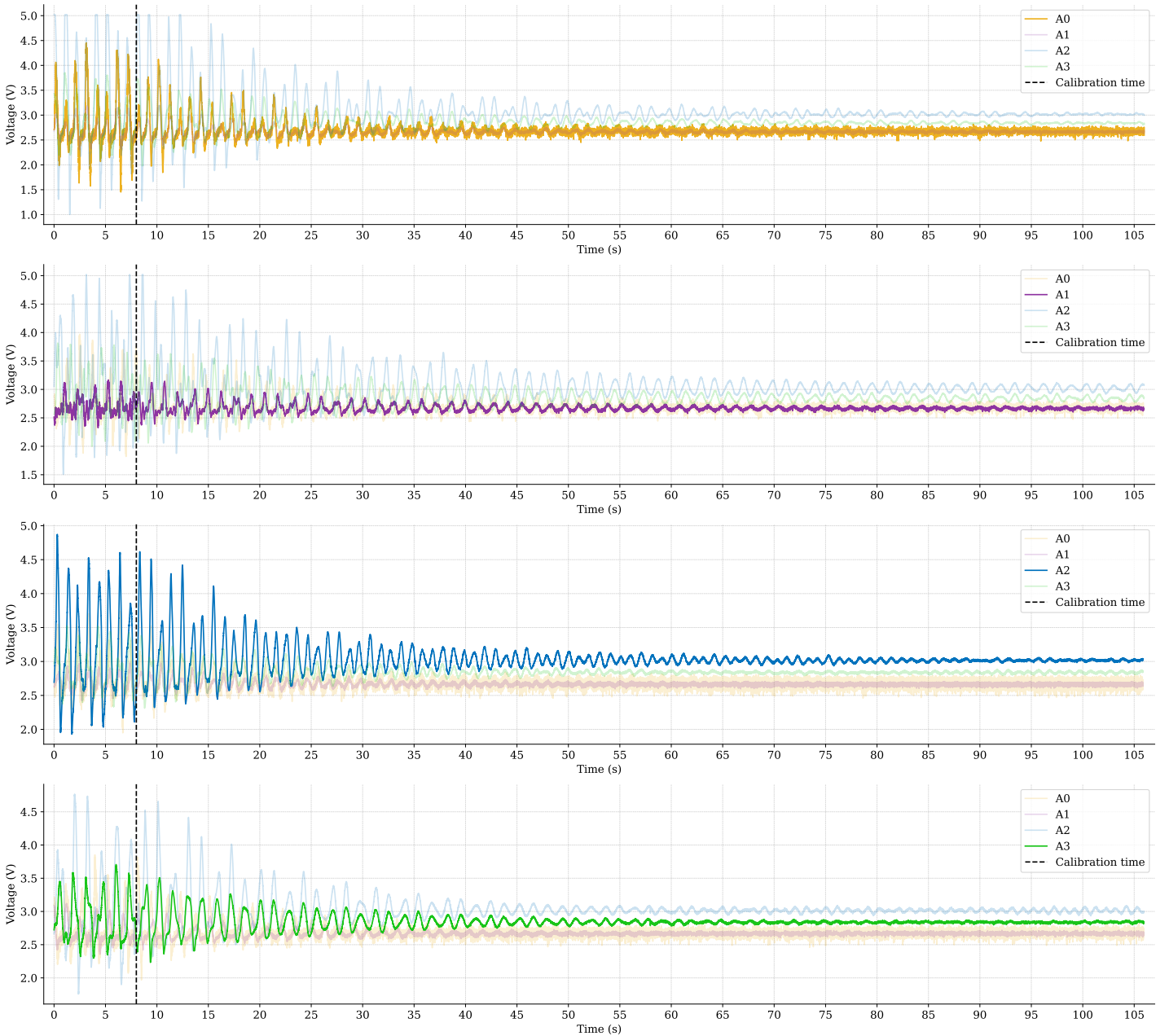


Figure 1: Stabilization of a single targeted OSEM channel with a damping time ≈ 70 s. The applied damping also couples to the other OSEM channels.

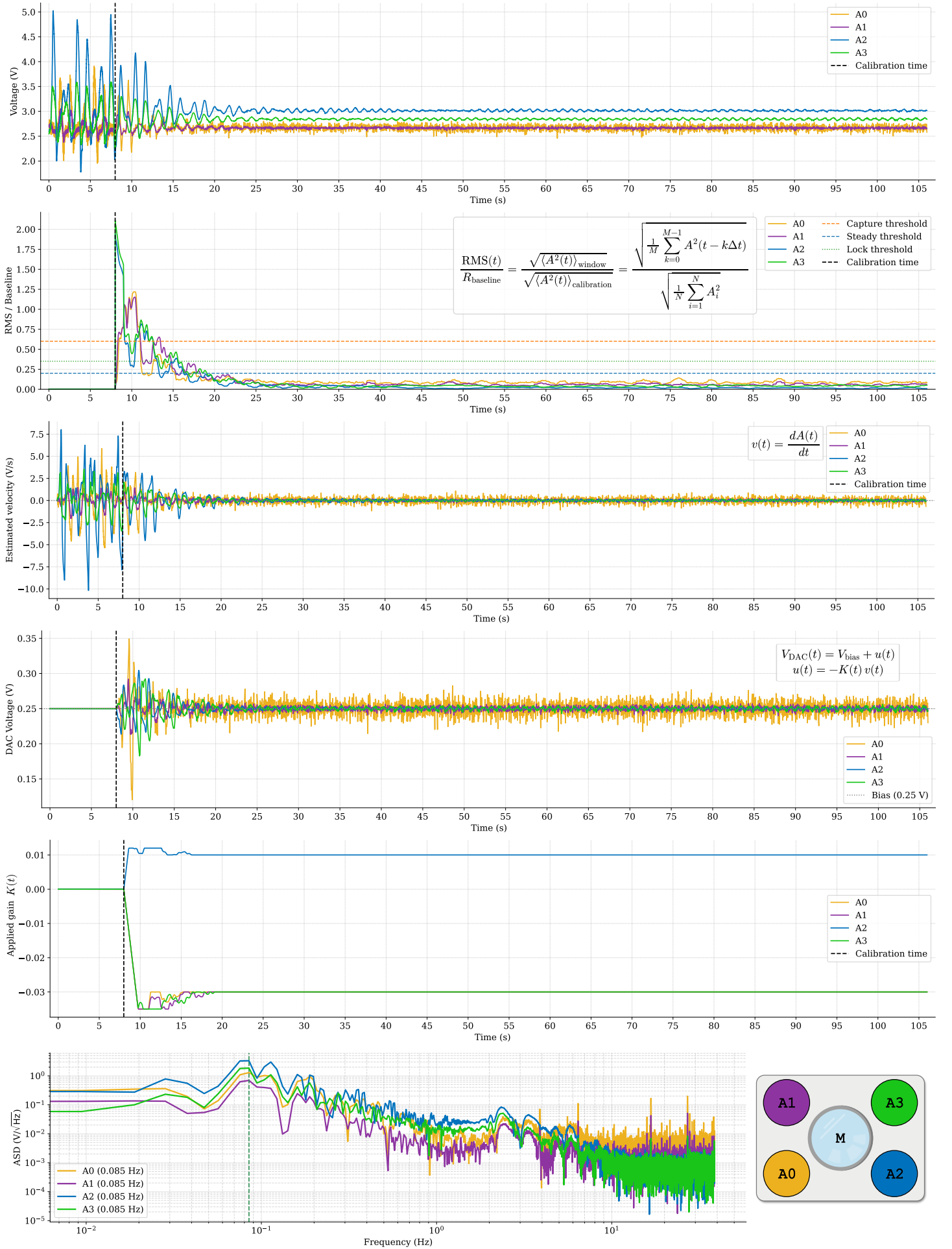


Figure 2: Simultaneous four OSEM channel stabilization. Compared to single-channel stabilization, damping time is significantly reduced to ≈ 17 s.

3. SINGLE OSEM CHANNEL STABILISATION

In an initial series of trials, only one OSEM/coil channel was actively damped at a time, with the remaining three left undriven and monitored passively. In every case the targeted channel was observed to damp out and settle to a stable, low-amplitude state, confirming that the velocity-feedback loop described above is sufficient to remove energy from the resonance it directly senses. Because all four OSEMs are mounted on, and therefore sense the motion of, the same rigid suspended optic, however, the four channels are not mechanically independent: they are different projections of a shared set of rigid-body normal modes. Actively damping only one of these projections still removes energy from the coupled structure as a whole, but does so far more slowly than a fully independent, per-mode actuation would, since the undriven channels can only lose energy indirectly, through their coupling to the one channel under active control. This is visible in Fig.1, where the actively targeted channel is plotted as a solid trace and the three undriven channels are shown faded: although only one channel is being driven, all four visibly decay over the course of the run, with the damping timescale of the whole system noticeably longer than that of the targeted channel alone.

4. SIMULTANEOUS FOUR OSEM CHANNEL STABILIZATION

Building on the observation that all four OSEM channels sample a common, coupled set of rigid-body modes, all four channels were subsequently damped simultaneously, each running its own independent velocity-feedback loop as described above. Under this configuration the entire optic was observed to settle to a stable, low-amplitude state in approximately 17 seconds – markedly faster than the single-channel case, where the same overall settling relied on the slow, indirect transfer of energy between modes through mechanical coupling alone. With all four channels actively damped, each rigid-body mode now loses energy directly through whichever sensor-actuator pair couples to it most strongly, rather than depending on a single channel to drain the whole coupled system; the four decentralized loops act in parallel rather than in series, so the overall damping rate is closer to the sum of the four individual loops' contributions than to the single slowest coupling path. It is worth noting that this remains four independent single-input single-output loops rather than a true multi-degree-of-freedom controller – each OSEM only ever drives its own coil – so the improvement comes from parallel coverage of the coupled modes rather than from any explicit decoupling between

them. Fig.2 shows a detailed breakdown of this run: the bandpassed signal, the applied gain, the scheduled gain-to-baseline ratio, and the raw sensor readings for all four channels across the full damping sequence.

In Fig.3, the detected oscillation peaks for each channel are shown together with the fitted exponential envelope. The fit quality, $R^2 \in [0.65, 0.92]$ (Table 1), is noticeably lower than in the clean single-channel case, indicating that the decay does not follow a pure exponential as closely here. This is expected rather than anomalous: as discussed in Section 3, the four OSEM channels are coupled projections of a shared set of rigid-body normal modes rather than independent single-degree-of-freedom oscillators, so a real decay curve can depart from the idealised single-exponential model – particularly toward the tail of the record, where the assumption of one dominant, cleanly separated mode is weakest. Even so, the stabilisation is still meaningfully characterised by a single effective damping coefficient γ and decay time $\tau = 1/\gamma$ per channel (Table 1), which capture the dominant rate at which oscillation energy is removed even where the underlying dynamics are not perfectly single-mode.

Table 1. Damping coefficient γ , decay time $\tau = 1/\gamma$, and fit quality R^2 .

Channel	γ (1/s)	τ (s)	R^2
A0	0.2197	4.5511	0.6477
A1	0.1821	5.4901	0.7960
A2	0.2796	3.5767	0.7526
A3	0.1982	5.0452	0.9185

5. PLAN FOR NEXT WEEK

1. To improve the stabilisation algorithm to achieve more accurate and stable OSEM signals.
2. To extend the microcontroller board to support an additional four OSEMs, which shall enable controlling the second suspended mirror.

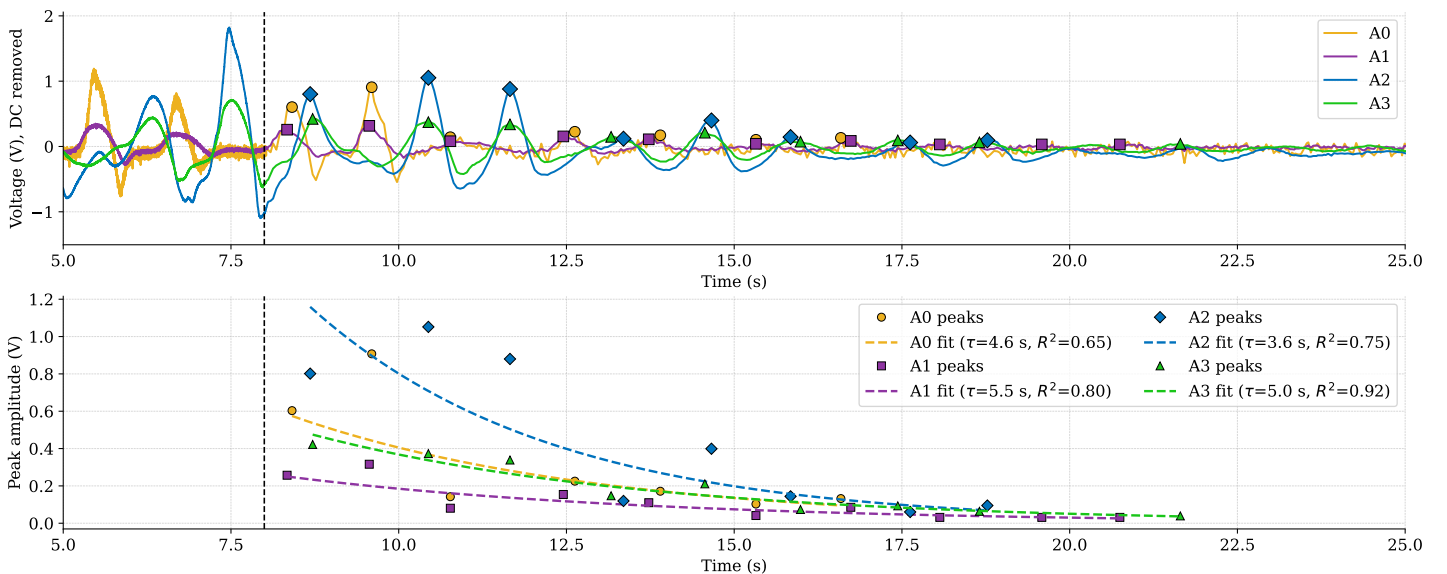


Figure 3: Peak detection (top) and exponential decay fit to the peak envelope (bottom) for each OSEM channel.